

Paper Title

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Abstract—

- **Problem:** improve settling prediction for a position-dependent dual gantry without surrendering the meaning of its physical parameters
- **Method:** combine a quasi-LPV first-principles baseline, dynamic parallel LFR augmentation, closed-loop rollout training, and state-level orthogonal projection
- **Evidence:** compare baseline, unconstrained augmentation, orthogonal variants, and a black-box model on held-out positions and motion profiles
- **Main result:** report predictive improvement together with the measured limits from short-horizon training, inactive added states, and non-orthogonal true missing dynamics
- **Conclusion:** state when the proposed decomposition is useful and which conditions prevent interpreting it as true parameter recovery

[TODO: Write the numerical abstract only after the final simulation and Telica result tables are frozen.]

Index Terms—grey-box identification, LPV systems, model augmentation, orthogonal projection, dual gantry

I. INTRODUCTION

A. Industrial motivation and modelling gap

- Settling prediction limits throughput in coupled high-precision gantries, so model error has a direct operational cost [1]
- The physics model exposes masses, stiffnesses, damping, and coupling, but omits position-dependent flexible dynamics that matter for prediction
- A black-box model offers flexibility but weakens extrapolation arguments and removes parameters useful for control design, digital twins, and health monitoring
- Grey-box dynamic augmentation is the target compromise: preserve the baseline and learn only the missing state dynamics [2]

[TODO: Which industrial benefit should lead: settling prediction, controller design, offline commissioning, or condition monitoring?]

B. Research question and scope

- Main question: how should a dual-gantry physics model be augmented to improve settling prediction while retaining physical parameter interpretability?
- Aspect 1 asks whether the payload-position scheduling should remain explicit in a quasi-LPV baseline rather than frozen LTI models

- Aspect 2 asks whether a dynamic parallel augmentation with additional states captures flexible cross-coupling absent from the baseline
- Aspect 3 asks what state-level orthogonality can and cannot guarantee during joint estimation
- Aspect 4 asks whether the model generalises across operating points and motion profiles relative to a black-box state-space model
- Scope separates demonstrated simulation evidence, planned real-machine evaluation, and untested extrapolation beyond the multisine amplitude range

[TODO: Does the approved research-plan wording need to remain verbatim, or may the final question reflect the closed-loop and OBC developments?]

C. Contributions and paper organisation

- Derive a compact continuous-time LPV-LFR realisation of the dual-gantry baseline with rational dependence on payload position
- Extend dynamic parallel LFR augmentation to a closed-loop, self-scheduled state-space setting with learned additional states
- Construct state-level orthogonality for the identifiable physical-parameter tangent space and quantify its true-parameter recovery condition
- Evaluate predictive fit, physical-parameter error, added-state contribution, generalisation, and black-box performance under matched data
- Paper organisation: baseline, augmentation, OBC, setup, results, discussion, conclusion, then supporting derivations

[TODO: Confirm whether references and appendices are excluded from the 12-page IEEE limit and whether TU/e requires front matter beyond the IEEE paper.]

II. DUAL-GANTRY SYSTEM AND PHYSICS BASELINE

A. System coordinates and equations of motion

- Define logical coordinates $q = [X, \Theta, Y]^T$, stage-force transformation, measured outputs, and the payload operating range [1]
- Present $M(Y)\ddot{q} + C\dot{q} + Kq = u$ and identify where payload position creates structural coupling and rational dependence
- Explain the free X and Y integrators caused by zero translational stiffness and why a small velocity bias accumulates into position error

- State which physical effects remain outside the baseline, including the hidden flexible mode and dry friction used in validation
- Verify $M(Y)$ is invertible throughout the operating range before any rollout or LFR transformation

[TODO: How much hardware description is needed to connect the logical coordinates to the two rails without consuming a full system figure?]

B. Quasi-LPV and LPV-LFR representation

- Treat measured payload position Y as the endogenous scheduling variable, making the model quasi-LPV [3]
- Keep the plant in continuous time and integrate with fixed-step RK4; explain the departure from the research plan's discrete-time ZOH route
- Factor the rational $M(Y)^{-1}$ dependence into an LFR with repeated scalar block $\Delta(Y) = YI$ [4], [5]
- Use structural SVD compression to reduce the scheduling loop from six to four channels without changing the input-state map
- State the LFR signal dimensions and name the interconnection variables consistently with standard lower-LFR terminology

[TODO: Ask Roland for the preferred name and notation for the compressed internal LFR dimensions currently described informally as n and m .]

C. Physical parameters and identifiable combinations

- Distinguish the fourteen raw parameters from the ten combinations identifiable through the gantry equations
- Optimise reduced positive parameter combinations and map them back to interpretable masses, stiffnesses, damping, geometry, and centre-of-mass terms
- Show the Jacobian from raw parameters to identifiable combinations; naive parameter merging is not a valid recovery argument
- Parameter recovery without augmentation is stated as a verification of the estimation pipeline [one sentence]
- Use relative combination error as the interpretability metric and disclose the special normalisation required when a combination crosses zero

[TODO: Should the raw-to-reduced Jacobian be printed in the appendix or shown only as rank and singular values?]

III. DYNAMIC LPV-LFR AUGMENTATION

A. Parallel augmentation with additional states

- Use a dynamic parallel LFR so the learned component adds missing dynamics while the physics state update remains explicit [2]
- Partition the model state into six physical states and learned additional states, with the ANN writing both partitions and the output reading physical states only
- Explain why dynamic states are needed for an omitted resonance and why static output correction or parameter re-estimation cannot create new poles

- Define the state router, ANN inputs, zero output readout at initialisation, and the recurrent path by which additional states affect later physical outputs
- The additional-state Jacobian blocks replace any claim that the learned states possess separate explicit A and B matrices

[TODO: How many added states can be justified from the omitted-mode physics rather than selected only by validation performance?]

B. Encoder and state initialisation

- Initialise each truncated rollout with an encoder using past input-output samples and a model-based linear map for the physical rows [6]
- Acknowledge that the published linear initialisation does not determine the additional-state rows, which begin from an unsupported random map
- Explain the zero-gradient path into added states at initialisation and the two zero gates that delay their use
- Use the exact simulated state for encoder-only diagnosis, but never as information available to the deployable model
- Compare encoder-history lengths and ANN capacity because the chosen windows determine reconstructability and startup transients

[TODO: Should the trained encoder remain part of the final model, or should a longer measured-history observer be treated as a separate deliverable?]

C. Closed-loop rollout objective

- Drive the model with $\hat{u}_k = u_k + C_{fb}(y_k - \hat{y}_k)$ so the simulated linear controller reacts to model error as the machine controller did
- Derive the residual form from controller linearity only and cite direct closed-loop augmentation as the published fallback for nonlinear controllers [7]
- Closed-loop training avoids reconstructing plant and controller states before every window because their prior response is already carried by recorded input
- Mark the departure from the research plan's open-loop free-run training and explain that feedback can hide model error and orthogonality overlap
- Train on short normalised windows with burn-in, but select and report on long free runs to expose integrated velocity bias

[TODO: Can ASMPT confirm saturation, anti-windup, and within-record controller scheduling, the three assumptions that determine whether the residual form is valid?]

D. Why augmentation rather than competing residual models

- Parameter refitting cannot represent a missing state and a static correction cannot reproduce flexible memory
- A residual FRF or fitted mass line is a useful linear benchmark but does not supply one nonlinear position-dependent state-space model for arbitrary trajectories

- Basis-function augmentation would improve transparency but requires choosing the missing physics class in advance and scales poorly across coupled effects
- Kessels supplies the closest closed-loop state-extension precedent, while this work adds LPV-LFR structure and an interpretability safeguard [7]
- The black-box comparator tests whether retaining the physics baseline adds value beyond model capacity alone

[TODO: Which competing method deserves an implemented benchmark beyond the required black-box model: residual BLA, basis functions, or parameter-only refitting?]

IV. INTERPRETABILITY BY ORTHOGONAL CONSTRUCTION

A. Negation and the protected tangent space

- Joint estimation is non-unique when the ANN can reproduce or cancel changes reachable by physical parameters [8], [9]
- Define interpretability as recovery of the ten identifiable physical combinations, not visual plausibility of individual raw parameters
- Stack one-step sensitivities $J = \partial f_\theta / \partial \theta$ over a data-derived reference set in normalised state coordinates
- Protect the rank-ten tangent space and compare the optional affine-offset column as a separate arm rather than silently changing the definition
- Report tangent-space rank, condition number, and correlated sensitivity columns before interpreting parameter recovery

[TODO: Should the thesis call the affine arm a stricter non-negation definition or only a sensitivity analysis around the chosen tangent definition?]

B. State-level orthogonal-by-construction map

- Project the ANN write into the physical rows with $I - JJ^+$, eliminating the penalty weight used by soft orthogonality regularisation [8]
- Freeze the reference basis within each epoch and refresh it as the physical parameters move, keeping the recurrent transition causal
- Differentiate through the projection coefficient while treating the stacked basis as constant during each optimisation interval
- Orthogonality applies to the physical one-step write; additional-state rows are orthogonal to the baseline by the baseline's zero extension
- Additional states still propagate into later physical outputs, so one-step state orthogonality does not imply trajectory-output orthogonality
- Separate the no-projection, ten-column tangent, and eleven-column affine arms under matched data, initialisation, and optimisation budgets

[TODO: Is one-step state orthogonality the intended thesis contribution, or must the trajectory-level penalty also appear despite its unresolved weight selection?]

C. Condition for true parameter recovery

- OBC makes the local decomposition unique, but true recovery additionally requires the real missing dynamics Δ^* to satisfy $J^\top \Delta^* = 0$
- Derive the least-squares parameter bias $\hat{\theta} - \theta^* = J^+ \Delta^*$ and define $\rho^* = \|JJ^+ \Delta^*\| / \|\Delta^*\|$
- Measure $\rho^* = 0.205$ for the hidden absorber, so orthogonality fails on the current excitation and zero parameter error is not the valid target
- The predicted combination-error floor is 4.96%, dominated by a mass-imbalance direction sharing the absorber's equilibrium offset
- Exact pointwise orthogonality is impossible for a nonzero force on this system; only excitation-averaged orthogonality can be designed

[TODO: Should the proposed orthogonal multisine design be a future-work result, or must it be run before claiming a constructive remedy?]

V. EXPERIMENTAL SETUP

A. Simulation benchmark and excitation

- Generate closed-loop records from the physics baseline plus a truth-only flexible absorber, retaining exact states solely for diagnostics
- Use multisine excitation that covers the absorber pole and anti-resonance while respecting position, velocity, yaw, and force limits
- Assign independent random multisine phases to training, validation, and test records so equal spectra do not duplicate trajectories
- Include multisine-free cycloidal motion profiles matching the measured machine motion to test the operational distribution shift
- Treat the Coulomb-friction benchmark as pending because the current friction and frictionless datasets have mismatched excitation amplitudes
- State the sample rates, record lengths, number of windows, and total independent motion time rather than quoting overlapping-window count as data volume

[TODO: Is the hidden linear absorber sufficient simulation complexity, or is a truth-only cubic spring required to demonstrate nonlinear augmentation?]

B. Real Telica data and split

- Use the supervisor-defined split of 11 training, two validation, and two test operating points at 20 kHz
- Convert rail positions and currents to logical metres and Newtons; form velocity states by central differences because direct velocity is not logged
- Keep all iterations at one operating point in the same split to prevent identical reference trajectories leaking across sets
- Describe the X and Y operating-point grid and show whether training covers the interpolation region occupied by both test points

- Use iteration 0 for controller checks and later ILC iterations for distinct feedforward conditions without treating them as independent trajectories
- Mark real-data training and held-out evaluation as planned until a final reproducible run and force-calibration convention exist

[**TODO:** Should direct drive velocity be requested, and should coverage be shown as a 2D position grid or a 3D state-input occupancy plot?]

C. Training protocol, comparators, and metrics

- Match records, seeds, windows, optimiser budget, and checkpoint criterion across baseline-only, unconstrained, tangent-OBC, affine-OBC, and black-box arms
- Compare short-horizon normalised RMS used for optimisation with 12 s free-run RMS, BFR, settling error, and per-channel spectra
- Report physical combination error, projection overlap, added-state ablations, and encoder-init sensitivity alongside predictive metrics
- Give the black-box model enough capacity and horizon to be a capability comparison, while disclosing its failed long-horizon training behaviour
- Evaluate interpolation across held-out payload positions, unseen multisine phase, multisine-free motion, and amplitude extrapolation separately
- Predeclare expectation, acceptance criterion, verdict, and mechanism for every result rather than presenting isolated scores

[**TODO:** Are multiple seeds or a Monte Carlo study required, and which uncertainty source should it sample: initialisation, noise, phase, or operating point?]

D. Reproducibility and implementation checks

- Verify controller reconstruction, coordinate transforms, normalisation, RK4 transition, and LFR equivalence before training
- Use the same controller, data conversion, and rollout entry point for training, validation, and checkpoint selection
- Record configuration, software precision, seed, dataset identifier, and checkpoint provenance for every reported arm
- Keep oracle states and truth parameters out of the trainable pipeline and label every diagnostic that uses them

[**TODO:** Which implementation checks belong in the main paper, and which should move to the appendix or repository documentation?]

VI. RESULTS

A. Baseline and closed-loop formulation

- Expectation and criterion: the LPV-LFR and original physics equations agree numerically across the payload range before model-error comparisons are meaningful
- Verdict and reason: report inertia invertibility, LFR/RK4 equivalence, and the remaining baseline error caused by the truth-only absorber

- Figure: baseline and oracle output errors versus time for representative payload positions; supports the claim that the omitted mode is systematic and position dependent
- Table: baseline free-run RMS or BFR for LPV and frozen-LTI models across held-out Y ; tests the research plan's Aspect 1 rather than assuming LPV superiority
- Expectation and criterion: direct and residual closed-loop forms match at numerical precision when controller assumptions hold
- Figure: recorded input, controller correction, and model error over one rollout; supports the interpretation that feedback can suppress visible output error while the open-loop model remains wrong

[**TODO:** The LPV-versus-frozen-LTI comparison is required by the plan but may not yet have a final matched result; run it or narrow Aspect 1 explicitly.]

B. Predictive benefit and additional-state contribution

- Expectation and criterion: augmentation must reduce held-out long-horizon error, not only the short training loss, relative to the same encoder-initialised baseline
- Verdict and reason: report the best validated improvement and the gap between window fit and free-run accuracy caused by integrated velocity bias
- Figure: validation RMS against optimiser updates for baseline and augmented arms, with window and 12 s metrics; supports or rejects objective alignment
- Figure: error spectra before and after augmentation with absorber pole and anti-resonance marked; tests whether the learned correction targets the missing band
- Figure: clamp each additional state and the full added-state vector to zero during rollout; quantifies what the trained latent dynamics add or harm
- Table: standard model, no-added-state ablation, zero-initialised latent path, and longer encoder history; connects structure changes to predictive benefit
- State the observed weak or harmful added-state contribution as a result if the final run does not reverse it, rather than inferring usefulness from state magnitude

[**TODO:** Are state Gramians useful for explaining each added state's contribution, or would intervention-based ablation be more direct and less linearisation-dependent?]

C. Parameter interpretability and orthogonality

- Expectation and criterion: OBC should reduce physical combination error without materially worsening held-out predictive fit relative to unconstrained joint estimation
- Verdict and reason: the completed arms reach 7.20% no projection, 7.69% tangent OBC, and 8.47% unfinished affine OBC from a common 9.49% start, so aggregate recovery does not establish OBC superiority
- Figure: validation fit versus combination error over training for all three arms; supports the claim that near-equal fit permits materially different physical parameters

- Figure: per-parameter error at matched budget; shows which combinations projection protects and which near-flat directions redistribute error
- Table: basis rank, condition, overlap, final and minimum parameter error, and validation RMS; links algorithm health to the recovery verdict
- Figure: ρ^* and predicted versus observed parameter bias by singular direction; supports the conditional result that true recovery is limited by non-orthogonal missing physics
- Explain why the 4.96% predicted floor accounts for part of the residual squared error but does not match the observed per-parameter pattern or prove convergence

[TODO: Should the incomplete affine arm be retained, rerun to a matched budget, or reported only as exploratory evidence?]

D. Generalisation and black-box comparison

- Expectation and criterion: improvement must persist on unseen positions, phases, multisine-free motion, and the real-machine test split without retuning
- Verdict and reason: separate completed simulation evidence from pending Telica results and explain any failure by excitation coverage, initialisation, or objective mismatch
- Figure: model error over the held-out cycloidal settling move for baseline, augmented model, and black box; supports the application claim on a machine-like trajectory
- Table: RMS, BFR, settling error, parameter availability, and training data for baseline, augmented, and black-box models on every held-out domain
- Figure: error versus multisine amplitude or state-input distance from the training set; tests extrapolation rather than calling interpolation generalisation
- Report the black box's early improvement and later free-run degradation as a negative comparator result if no stable matched run supersedes it
- Parameter recovery trajectories from the synthetic verification study are contextual only [one sentence]

[TODO: What is the minimum final evidence needed before claiming generalisation: Telica test points, multisine-free simulation, amplitude extrapolation, or all three?]

VII. DISCUSSION

A. Answers to the four research aspects

- Aspect 1: state whether explicit Y scheduling earns measurable predictive value over frozen LTI baselines
- Aspect 2: distinguish the expressive value of dynamic augmentation from the observed trainability of its added states
- Aspect 3: conclude that OBC enforces a unique local split but true parameter recovery is conditional on the real discrepancy's orthogonality and optimiser convergence
- Aspect 4: compare simulation interpolation, motion-profile transfer, amplitude extrapolation, real-data evidence, and the black-box result without merging them into one score

- Interpretability matters when physical parameters support controller redesign or health monitoring, but its practical value must be tied to a named downstream decision

[TODO: Which result is the primary answer if predictive accuracy and parameter recovery favour different arms?]

B. Limitations and lessons

- Closed-loop feedback improves numerical trainability but can hide plant mismatch, suppress orthogonality overlap, and reward a model that is poor in open loop
- Short windows underweight velocity DC bias on free integrators; varying encoder and rollout length is therefore a structural sensitivity test, not routine tuning
- One absorber, noiseless simulation, one main seed, and an incomplete friction benchmark limit claims about nonlinear, noisy, and dissipative missing dynamics
- Central-difference velocities, force calibration, and the train-validation-test coverage grid bound the credibility of any real-data conclusion
- The research plan expected validated real-machine generalisation; any missing Telica result is a departure to disclose, not replace with design documentation
- Augmentation versus residual-FRF fitting remains a trade between trajectory-domain nonlinear prediction and a simpler frequency-domain account of missing dynamics

[TODO: Should a cubic-spring benchmark, noise study, and Monte Carlo analysis be completed, or should the thesis explicitly narrow its claims to the measured linear absorber case?]

C. Implications and future work

- Design excitation so $J^\top \Delta^* = 0$ when true parameter recovery, rather than only a unique decomposition, is required
- Separate learning rates for physical and neural parameters and compare convergence against the predicted nonzero OBC bias floor
- Replace the short-horizon objective or couple windows so integrated velocity bias is visible during training
- Repair and rerun the Coulomb benchmark before claiming that the method learns a legitimate nonzero-mean residual
- Extend evaluation beyond the multisine range and use machine-relevant trajectories to delimit safe interpolation and extrapolation

[TODO: Which future-work item is a prerequisite for the thesis claim and which genuinely belongs after graduation?]

VIII. CONCLUSION

A. Conclusions

- Answer whether LPV-LFR dynamic augmentation improves dual-gantry settling prediction over the physics baseline and black-box comparator
- State whether the learned additional states contributed measurable missing dynamics under the final validated configuration

- Conclude that state-level OBC prevents local tangent overlap by construction, while true parameters are recovered only when the physical discrepancy is orthogonal on the chosen excitation
- Give the final quantitative trade between prediction error and physical combination error on held-out data
- Close with the practical recommendation for when ASMPT should use the augmented model and when the baseline or a simpler residual model is safer

[TODO: Rewrite every conclusion bullet after freezing the final result table; no planned or diagnostic result may appear as achieved.]

APPENDIX A

SUPPORTING DERIVATIONS AND REPRODUCIBILITY

A. Baseline and LFR details

- Give the logical-to-stage transformation, expanded inertia matrix, and proof of invertibility over the operating range
- List the compressed LPV-LFR matrices and internal signal dimensions needed to reproduce the baseline transition
- Provide the raw-to-identifiable parameter map and its Jacobian rank

[TODO: Move these items into supplementary material if appendices count toward the 12-page limit.]

B. OBC and experiment details

- Derive the stacked projector, rank-deficient pseudoinverse case, epoch refresh, and additional-state zero extension
- Derive $J^+ \Delta^*$ and list the measured singular-direction contributions to the predicted parameter bias
- List dataset identifiers, split membership, hyperparameters, seed, and exact metric definitions for every reported arm

[TODO: Can code and data tables be cited through a repository or DOI instead of spending paper space on full reproducibility metadata?]

REFERENCES

- [1] "Source pending verification," citation metadata not yet verified.
- [2] "Source pending verification," citation metadata not yet verified.
- [3] "Source pending verification," citation metadata not yet verified.
- [4] "Source pending verification," citation metadata not yet verified.
- [5] "Source pending verification," citation metadata not yet verified.
- [6] "Source pending verification," citation metadata not yet verified.
- [7] "Source pending verification," citation metadata not yet verified.
- [8] "Source pending verification," citation metadata not yet verified.
- [9] "Source pending verification," citation metadata not yet verified.